

EDUARDO GARCIA

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EDUCATION

The University of Texas at Tyler

Bachelor of Science in Mechanical Engineering | Houston TX

Honors: Dean's List (Fall 2023)

Relevant Coursework: Finite Element Analysis, Dynamics of Machinery, Mechatronics, Mechanics of Materials, Thermodynamics, Machine Design, Fluid Mechanics, Heat Transfer

Expected Spring 2027

Houston Community College

Associate of Science in Engineering Science | Houston TX

Completed Fall 2024

ENGINEERING PROJECTS

Skateboard Deck FEA — Static Structural Analysis

University of Texas at Tyler | Finite Element Analysis | ANSYS Workbench, SolidWorks

Spring 2026

- Modeled a seven-ply skateboard deck in ANSYS Workbench with orthotropic maple and CFRP material assignments; performed a three-stage static FEA comparing flat, curved, and CFRP configurations under a 5,000 N quasi-static load.
- Achieved a 48% deformation reduction in the CFRP laminate versus flat maple baseline (Factor of Safety 26.5 versus 0.87); validated results against Euler-Bernoulli beam theory through a mesh convergence study across six refinement levels.

V90 Twin-Cylinder Pump — Kinematic SVAJ Design

University of Texas at Tyler | Dynamics of Machinery | MATLAB, SolidWorks Motion

Spring 2026

- Parametrically swept 1,240 slider-crank linkage geometries in MATLAB to optimize a V90 twin-cylinder pump for 10 L/s flow at 10 bar; filtered 291 feasible candidates using a weighted kinematic composite score.
- Final design reduced peak piston acceleration by 76% and velocity by 52% versus baseline; validated kinematics in SolidWorks Motion within 0.2% velocity agreement and confirmed V90 flow pulsation advantages.

Color Sorting Robotic Arm — Mechatronics

University of Texas at Tyler | Mechatronics | Arduino Uno, C++, 3D Printing

Spring 2026

- Designed and built a four degree of freedom Arduino-based robotic arm integrating a TCS34725 RGB color sensor, HC-SR04 ultrasonic sensor, and LDR for autonomous detection, pickup, and sorting of red and green objects.
- Implemented closed-loop proportional control in C++ to suppress servo jitter; 3D printed structural components and resolved mechanical slippage through iterative design refinement.

NASA L'SPACE Mission Concept Academy

Multidisciplinary STEM Team | Siemens NX

Fall 2023

- Designed and analyzed a CubeSat structural frame for an Enceladus astrobiology exploration mission using Siemens NX; collaborated with a multidisciplinary team on material selection and structural requirements for deep space conditions.

WORK EXPERIENCE

Rice University, Research Experience for Undergraduates

Undergraduate Researcher, Houston TX

April 2023 to July 2023

- Developed carbon nanotube film-based thermoelectric generators; designed and assembled a custom measurement device to characterize Seebeck coefficient, electrical conductivity, and thermal conductivity.
- Analyzed experimental data in MATLAB and Python to validate results against theoretical models and published literature.

Walmart Supercenter

Team Associate, Houston TX

November 2020 to February 2022

- Cross-trained across stocking, product facing, and backroom logistics to keep merchandise available and the sales floor organized during peak hours.
- Maintained reliable attendance across 15 months and stepped in for additional shifts when the team needed coverage.

H-E-B

Department Assistant, Meat Department, Houston TX

April 2019 to January 2020

- Assisted customers with product selection and preparation guidance in a specialized, high-demand department; communicated daily in both English and Spanish with a diverse customer base.

TECHNICAL SKILLS

CAD and Simulation: SolidWorks, AutoCAD, Fusion 360, Siemens NX, ANSYS Workbench (Static Structural, FEA), SolidWorks Motion, 3D Printing

Programming and Analysis: MATLAB, Python, C++ (Arduino IDE), Microsoft Excel

Other: Technical Writing and Documentation, Engineering Design and Validation

Languages: English (Fluent), Spanish (Fluent)

ORGANIZATIONS AND HONORS

American Society of Mechanical Engineers (ASME), *Student Member*

2024 to Present

Society of Hispanic Professional Engineers (SHPE), *Student Member*

2023 to Present

American Institute of Aeronautics and Astronautics (AIAA), *Student Member*

2023 to Present